New Lower Bounds for C_4-Free Subgraphs of the Hypercubes Q_6, Q_7, and Q_8: Constructions, Structure, and Computational Method

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Abstract

We establish new lower bounds $\mathrm{ex}(Q_7, C_4) \geq 304$ and $\mathrm{ex}(Q_8, C_4) \geq 680$ for the maximum number of edges in a C_4-free subgraph of the 7- and 8-dimensional hypercubes, and give a modern computational reproduction of $\mathrm{ex}(Q_6, C_4) = 132$. All bounds are witnessed by explicit constructions supplied as edge lists with machine-verifiable certificates: a complete, deterministic enumeration of all four-cycles (240 for Q_6, 672 for Q_7, 1 792 for Q_8) establishes C_4-freeness, and each certificate is reproducible by an independent, dependency-free verifier.

For Q_7 we identify 19 866 distinct C_4-free subgraphs on 304 edges; their *dimension profiles* fall into exactly 20 types. All 19 866 solutions share a rigid structural core: degree sequence $\{4^{32}, 5^{96}\}$, spectral radius $\lambda_1 \approx 4.787$, and local maximality. Pairwise Hamming distances range from 36 to 260. Whether these solutions exhaust all 304-edge C_4-free subgraphs of Q_7 remains open.

For Q_8 we analyse the local structure of the 680-edge construction: every non-edge of the construction creates at least one C_4, and 1 076 independent searches at 681 edges did not achieve zero violations. These observations constitute computational evidence, not a proof, of the conjectured equality $\mathrm{ex}(Q_8, C_4) = 680$.

The constructions are found by a two-phase simulated annealing algorithm with $\mathrm{Aut}(Q_n)$-based diversification, described in full detail. For Q_6 we additionally provide an ILP-based proof that $\mathrm{ex}(Q_6, C_4) \leq 132$, reproducing the exact value computationally.

Edge lists, ILP files, and source code are publicly available at https://github.com/minamominamoto/c4free-hypercube.

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Keywords: hypercube, C_4-free, extremal graph theory, simulated annealing, integer linear programming, dimension profile.

Contents

1 Introduction

The n-dimensional hypercube Q_n has vertex set $\{0,1\}^n$, two vertices adjacent iff they differ in exactly one coordinate; it has 2^n vertices and $n \cdot 2^{n-1}$ edges. The problem of determining $\mathrm{ex}(Q_n, C_4) = \max\{|E(G)| : G \subseteq Q_n,\ G \text{ is } C_4\text{-free}\}$ was raised by Erdős [6] and remains open in general.

The best known asymptotic bounds are

$$\tfrac{1}{2}(n + \sqrt{n}) \cdot 2^{n-1} \ \le\ \mathrm{ex}(Q_n, C_4) \ \le\ 0.6032 \cdot n \cdot 2^{n-1}, \tag{1}$$

where the lower bound (valid for $n = 4^r$) is due to Brass, Harborth, and Nienborg [3], and the upper bound to Baber [1] via the flag algebra method, improving the earlier bound of Thomason and Wagner [8]. Balogh et al. [2] give 0.6068 via a related method. For general $n \ge 9$, the weaker bound $\frac{1}{2}(n + 0.9\sqrt{n}) \cdot 2^{n-1}$ applies [3]. Exact values have been determined for small n; Harborth and Nienborg [7] proved $\mathrm{ex}(Q_6, C_4) = 132$.

Our main results are:

Theorem 1. *(i)* $\mathrm{ex}(Q_7, C_4) \ge 304$.

(ii) $\mathrm{ex}(Q_8, C_4) \ge 680$.

Both bounds are realised by explicit constructions (supplementary files `q7_edges_304.jsonl` and `q8_edges_680.jsonl`).

The paper is organised as follows. Section 2 describes the C_4 enumeration and verification procedure used throughout. Sections 3–4 present the Q_7 and Q_8 constructions. Section 5 classifies the 19 866 Q_7 solutions found by their dimension profiles, yielding exactly 20 types. Section 6 describes the two-phase simulated annealing algorithm. Section 7 reproduces the exact value $\mathrm{ex}(Q_6, C_4) = 132$ via ILP. Section 8 lists open problems.

Table 1: Known values and new lower bounds of $\mathrm{ex}(Q_n, C_4)$. The BHN column gives the Brass–Harborth–Nienborg general estimate.

n	$\lvert E(Q_n)\rvert$	$\mathrm{ex}(Q_n, C_4)$	BHN est.	Source
4	32	24	≈ 25.6	Chung [4]
5	80	56	≈ 51.2	Emamy-K et al. [5]
6	192	132	≈ 96.0	Harborth–Nienborg [7] (reproduced)
7	448	$\geq$ **304**	≈ 300.2	This work
8	1024	$\geq$ **680**	≈ 674.9	This work

2 C_4 Enumeration and Verification

Lemma 2 (C_4 enumeration in Q_n)**.** *Every four-cycle of Q_n is uniquely determined by a pair of coordinate directions $0 \leq i < j \leq n-1$ and a base vertex $b \in \{0,1\}^n$ with $b_i = b_j = 0$. Its four vertices are*

$$b, \quad b \oplus e_i, \quad b \oplus e_j, \quad b \oplus e_i \oplus e_j,$$

and the total number of four-cycles in Q_n is $\binom{n}{2} \cdot 2^{n-2}$. For $n = 6$: 240; for $n = 7$: 672; for $n = 8$: 1 792.

Verification algorithm. Given a candidate edge set $E \subseteq E(Q_n)$ stored as a hash set, the following procedure performs a complete, deterministic C_4-freeness check:

for each pair (i, j) with $0 \leq i < j \leq n-1$:
for each $b \in \{0,1\}^n$ with $b_i = b_j = 0$:
let $v_1 = b,\ v_2 = b \oplus e_i,\ v_3 = b \oplus e_j,\ v_4 = b \oplus e_i \oplus e_j$
if $\{v_1v_2,\ v_1v_3,\ v_2v_4,\ v_3v_4\} \subseteq E$: **return** VIOLATION
return C4-FREE

We applied this algorithm to all constructions and all 19 866 solutions of Q_7; every call returned C4-FREE.

3 The Q_7 Construction

We constructed a C_4-free subgraph of Q_7 on 304 edges by penalty-based simulated annealing (Section 6), then certified it by Lemma 2's algorithm.

Proposition 3. *The 304-edge construction has the following properties:*

(i) Degree sequence: 32 vertices of degree 4 and 96 of degree 5 (degree sum $608 = 2 \cdot 304$).

(ii) Bipartite: $\lambda_1 \approx 4.787$, $\lambda_n \approx -4.787$, $\lambda_1 + \lambda_n = 0$ to machine precision.

(iii) $\mathrm{Tr}(A^4) = 5216$, attaining the C_4-free trace equality $\sum_i \deg(i)^2 + \sum_{uv \in E}(\deg u + \deg v) - 2|E|$.

(iv) Local maximality: every non-edge of Q_7 creates a C_4 when added.

Through 19 866 independent runs (Section 5.3) we found 19 866 C_4-free subgraphs of Q_7 on 304 edges with pairwise distinct edge sets (not quotiented by $\mathrm{Aut}(Q_7)$), all sharing the structural properties stated in Proposition 9.

4 The Q_8 Construction

Simulated annealing produced a C_4-free subgraph of Q_8 on 680 edges, certified by exhaustive enumeration of all 1 792 four-cycles.

Proposition 4. *The* 680*-edge* C_4*-free subgraph of* Q_8 *has the following properties:*

(i) Degree sequence $\{4^8, 5^{160}, 6^{88}\}$*; degree sum* $1360 = 2 \cdot 680$*.*

(ii) Bipartite, with $\lambda_1 + \lambda_n = 0$*.*

(iii) Local maximality: every non-edge of Q_8 *creates a* C_4 *upon addition.*

(iv) Edge density $680/1024 = 66.4\%$*, compared to* $304/448 = 67.9\%$ *for the* Q_7 *construction.*

Remark 5. Since 8 is not a power of 4, the exact BHN formula does not apply for $n = 8$. The applicable estimate is $\frac{1}{2}(8 + 0.9\sqrt{8}) \cdot 128 \approx 674.9$, which our 680-edge construction surpasses by 5.1 edges.

4.1 Local optimality of the 680-edge construction

For the 680-edge solution, every non-edge of Q_8 creates at least one C_4 upon addition. Among the 344 non-edges: exactly 1 creates exactly 1 new C_4; exactly 1 creates exactly 2; the remaining 342 create 3 or more.

Over 1 076 independent simulated-annealing trials targeting 681 edges, the minimum C_4 violation count observed was 1 (never 0), where the violation count denotes the number of four-cycles fully contained in the candidate edge set.

Conjecture 6. $\mathrm{ex}(Q_8, C_4) = 680$.

Table 2 compares the Q_7 and Q_8 constructions.

Table 2: Structural comparison of the Q_7 and Q_8 constructions.

Parameter	Q_7 (304 edges)	Q_8 (680 edges)
Total edges $\lvert E(Q_n)\rvert$	448	1024
Achieved edges	304	680
Edge density	67.9%	66.4%
Min degree	4	4
Max degree	5	6
Degree sequence	$\{4^{32}, 5^{96}\}$	$\{4^8, 5^{160}, 6^{88}\}$
BHN estimate	≈ 300.2	≈ 674.9
Improvement	$+3.8$	$+5.1$

5 Structural Classification of Q_7 Solutions

5.1 Dimension profiles and the twenty types

Definition 7 (Dimension profile)**.** For $G \subseteq Q_7$, let $e_i(G) = |\{uv \in E(G) : u \oplus v = 2^i\}|$. The *dimension profile* is the sorted tuple $\pi(G) = (e_{\sigma(0)} \geq \cdots \geq e_{\sigma(6)})$.

The dimension profile is invariant under coordinate permutations (a subgroup of $\mathrm{Aut}(Q_7)$ of order $7! = 5040$), giving a coarse but computable classification.

Proposition 8. *Among the* 19 866 *solutions, the dimension profile takes exactly* **20** *distinct values.*

Table 3 lists all 20 types with frequencies and structural invariants.

Table 3: The 20 dimension-profile types of 304-edge C_4-free subgraphs of Q_7, sorted by decreasing frequency. H: Shannon entropy of normalised profile (bits).

Rank	Profile π	Solutions	H	λ_1 (mean)
1	$[44, 44, 44, 44, 43, 43, 42]$	3155	2.8072	4.787
2	$[45, 45, 45, 43, 43, 42, 41]$	2913	2.8065	4.788
3	$[45, 45, 45, 43, 43, 43, 40]$	2200	2.8063	4.786
4	$[44, 44, 44, 44, 44, 43, 41]$	2116	2.8069	4.787
5	$[46, 46, 46, 42, 42, 42, 40]$	1756	2.8053	4.788
6	$[46, 46, 46, 42, 42, 41, 41]$	1181	2.8054	4.789
7	$[44, 44, 44, 44, 44, 44, 40]$	1085	2.8066	4.787
8	$[45, 45, 45, 43, 42, 42, 42]$	1064	2.8066	4.787
9	$[45, 45, 44, 43, 43, 43, 41]$	974	2.8067	4.787
10	$[44, 44, 44, 44, 44, 42, 42]$	931	2.8070	4.787
11	$[47, 47, 47, 41, 41, 41, 40]$	902	2.8037	4.789
12	$[45, 45, 43, 43, 43, 43, 42]$	439	2.8069	4.787
13	$[45, 44, 44, 43, 43, 43, 42]$	307	2.8070	4.787
14	$[46, 45, 45, 42, 42, 42, 42]$	236	2.8063	4.787
15	$[44, 44, 44, 43, 43, 43, 43]$	163	2.8073	4.787
16	$[47, 47, 44, 43, 41, 41, 41]$	153	2.8050	4.788
17	$[46, 46, 44, 42, 42, 42, 42]$	107	2.8062	4.787
18	$[48, 48, 48, 40, 40, 40, 40]$	101	2.8014	4.789
19	$[46, 46, 43, 43, 42, 42, 42]$	44	2.8063	4.787
20	$[46, 46, 45, 42, 42, 42, 41]$	39	2.8058	4.787
	Total	**19866**		

Observations: (1) λ_1 lies in $[4.786, 4.789]$ across all 20 types. (2) H ranges from 2.801 to 2.807, all near $\log_2 7 \approx 2.807$. (3) Type 18 ($[48, 48, 48, 40, 40, 40, 40]$) has 3-fold symmetry.

5.2 Shared structural properties

Proposition 9. *Every solution in the* 19 866 *satisfies:*

(i) Degree sequence $\{4^{32}, 5^{96}\}$*.*

(ii) $\lambda_1 + \lambda_n = 0$ *(bipartite),* $\lambda_1 \approx 4.787$*.*

(iii) $\mathrm{Tr}(A^4) = 5216$ *(C_4-free trace equality).*

(iv) Local maximality: no edge of Q_7 can be added without creating a C_4.

(v) Every edge of Q_7 appears in at least one solution (verified by taking the union of the edge sets over all 19 866 solutions).

5.3 Solution landscape

Pairwise Hamming distances $d_H(G, G') = |E(G) \triangle E(G')|$ over a random sample of 5 000 pairs range from 36 to 260 (mean ≈ 195.4, median 196).

Over 1 076 independent trials targeting 305 edges, the minimum C_4 violation count was 2 (never 0).

Conjecture 10. $\mathrm{ex}(Q_7, C_4) = 304$.

6 Computational Method

6.1 Two-phase simulated annealing

Phase 1 (Penalty SA). We minimise $f(E) = -|E| + \lambda V$ where V is the number of C_4 violations and $\lambda \in [0.30, 0.90]$. Temperature schedule: geometric from $T_0 \in [0.20, 4.00]$ to $T_1 \in [0.001, 0.030]$ over 3–15 million steps. A move toggles a single edge; the change in f is computed incrementally in $O(\deg)$ time using precomputed C_4-to-edge incidence lists.
Phase 2 (Swap SA). Edge count is fixed; swap moves (remove one edge, add one non-edge) minimise V. Parameters: up to 2×10^8 steps for Q_7 at 305 edges; 5×10^7 for Q_8 at 681 edges.
Diversification. Before each trial, a random element of $\mathrm{Aut}(Q_n)$ (random bit permutation composed with random bit flip) is applied to the current best solution. This prevents repeated exploration of the same automorphism class.

6.2 Comparison with known lower bounds

$$\begin{aligned} n = 7: &\quad \tfrac{1}{2}(7 + 0.9\sqrt{7}) \cdot 64 \approx 300.2, \quad \text{our bound: } 304\ (\Delta = +3.8,\ +1.25\%), \\ n = 8: &\quad \tfrac{1}{2}(8 + 0.9\sqrt{8}) \cdot 128 \approx 674.9, \quad \text{our bound: } 680\ (\Delta = +5.1,\ +0.75\%). \end{aligned}$$

6.3 Regularity trend across dimensions

Degree sequences across dimensions suggest increasing regularity: Q_6: $\{4^{32}, 5^{32}\}$; Q_7: $\{4^{32}, 5^{96}\}$; Q_8: $\{4^8, 5^{160}, 6^{88}\}$. The degree support $\{n-3, n-2, n-1\}$ may reflect an underlying structural principle for optimal C_4-free subgraphs of Q_n.

7 Computational Proof that $\mathrm{ex}(Q_6, C_4) = 132$

Harborth and Nienborg [7] proved $\mathrm{ex}(Q_6, C_4) = 132$ by combinatorial arguments. We reproduce this result computationally, providing an independent verification and a template for small hypercubes.

7.1 Lower bound: explicit construction

Simulated annealing (Section 6) finds a C_4-free subgraph of Q_6 on 132 edges, certified by exhaustive enumeration of all 240 four-cycles. The explicit edge list is provided in `q6_edges_132.jsonl`.

7.2 Upper bound: integer linear program

Label the 192 edges of Q_6 as $x_1, \ldots, x_{192} \in \{0, 1\}$. For each of the 240 four-cycles $\{e_1, e_2, e_3, e_4\}$, the constraint $x_{e_1} + x_{e_2} + x_{e_3} + x_{e_4} \leq 3$ ensures C_4-freeness. The ILP is:

$$\max \sum_{i=1}^{192} x_i \quad \text{subject to} \quad \sum_{e \in C} x_e \leq 3 \ \ \forall C \in \mathcal{C}_6, \quad x_i \in \{0, 1\}.$$

This formulation has 192 binary variables and 240 constraints. The MPS file `q6_ilp.mps` is provided for independent verification.

Proposition 11. *The optimal value of the above ILP is* 132*, establishing* $\mathrm{ex}(Q_6, C_4) \leq 132$*. Combined with the lower bound construction,* $\mathrm{ex}(Q_6, C_4) = 132$*.*

On standard desktop hardware, standard MIP solvers such as SCIP, HiGHS, or Gurobi verify this instance quickly.

8 Open Problems

1. Prove or disprove $\mathrm{ex}(Q_7, C_4) = 304$ (Conjecture 10).
2. Determine $\mathrm{ex}(Q_8, C_4)$ exactly (Conjecture 6).
3. What is the number of $\mathrm{Aut}(Q_7)$-orbits among all 304-edge C_4-free subgraphs of Q_7?
4. Is the degree sequence $\{4^{32}, 5^{96}\}$ necessary for any locally maximal C_4-free subgraph of Q_7 on 304 edges?
5. Can the flag algebra method of [1, 2] yield tight bounds for small $n \leq 8$?
6. Does the degree sequence pattern $\{(n-3)^a, (n-2)^b, (n-1)^c\}$ continue for $n = 9, 10, \ldots$?

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